



Explain Logistic Regression.
What is Logistic Response
Function and

Logit in Data Science

Logistic Regression (Exam-Oriented Answer)

Definition:

Logistic Regression is a **supervised machine learning algorithm** used for **classification problems**, especially when the dependent variable is **binary** (0 or 1, Yes/No, True/False).

Unlike linear regression, it predicts **probability** instead of continuous values.

1. Logistic Regression Model



Instead of fitting a straight line, logistic regression uses a **sigmoid (S-shaped) curve** to



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$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

linear regression, it predicts instead of continuous values.

1. Logistic Regression Model

Instead of fitting a straight line, logistic regression uses a **sigmoid (S-shaped) curve** to map predicted values between 0 and 1.

$$P(Y = 1|X) = \frac{1}{1+e^{-(\beta_0+\beta_1 X)}}$$

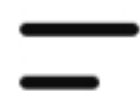
Where:

- $P(Y = 1|X)$ = Probability of class 1
- β_0, β_1 = Model parameters
- e = Euler's number

2. Logistic Response Function (Sigmoid Function)

The **logistic response function** transforms any real-valued number into a value between 0 and 1.

$$\sigma(z) = \frac{1}{1+e^{-z}}$$



2. Logistic Response Function (Sigmoid Function)

The **logistic response function** transforms any real-valued number into a value between **0 and 1**.

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

Where:

- $z = \beta_0 + \beta_1 X$

Key Properties:

- Output range: (0,1)
- S-shaped curve
- Used to interpret outputs as probabilities

Interpretation:

- If output $\geq 0.5 \rightarrow$  1
- If output $< 0.5 \rightarrow$ Class 0



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Interpretation:

- If output $\geq 0.5 \rightarrow$ Class 1
- If output $< 0.5 \rightarrow$ Class 0

3. Logit Function

The **logit** is the **logarithm of odds** (log-odds).
It transforms probability into a linear form.



$$\text{logit}(p) = \ln \left(\frac{p}{1-p} \right)$$



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- p = probability of success

=

• If output $< 0.5 \rightarrow$ Class 0

3. Logit Function

The **logit** is the **logarithm of odds** (log-odds).
It transforms probability into a linear form.

$$\text{logit}(p) = \ln \left(\frac{p}{1-p} \right)$$

Where:

- p = probability of success

Expanded form in Logistic Regression:

$$\ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X$$

4. Relationship Between Logistic Function and Logit

- **Logistic Function**  Converts linear output $\rightarrow$ probability



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So, they are **inverse** of each other.

=

ended form in Logistic Regres



$$\ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X$$

4. Relationship Between Logistic Function and Logit

- **Logistic Function:** Converts linear output → probability
- **Logit Function:** Converts probability → linear relationship

So, they are **inverse of each other**.

5. Example

Suppose we predict whether a student will pass (1) or fail (0):

- Input: Study hours
- Output: Probability of passing

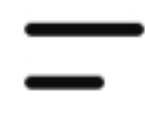


If m



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 $P = 0.8$



5. Example

Suppose we predict whether a student will pass (1) or fail (0):

- Input: Study hours
- Output: Probability of passing

If model gives:

$$P = 0.8$$

Logit:

$$\ln(0.8/0.2) = \ln(4)$$

6. Applications of Logistic Regression

- Spam email detection
- Disease prediction (diabetes, cancer)
- Credit risk analysis



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6. Applications of Logistic Regression

- Spam email detection
- Disease prediction (diabetes, cancer)
- Credit risk analysis
- Customer churn prediction

7. Advantages

- Simple and easy to implement
- Provides probability output
- Works well for binary classification

8. Limitations



- Costly to train



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- Sensitive to outliers

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- Provides probability output
- Works well for binary classification

8. Limitations

- Cannot handle complex non-linear relationships well
- Sensitive to outliers
- Assumes linearity in logit

Conclusion (For Exam Ending)

Logistic Regression is a powerful classification technique that uses the **logistic response function** to map outputs into probabilities and the **logit function** to establish a linear relationship between predictors and the **log odds** of the outcome.



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How are predicted values
obtained from Logistic
Regression? Explain

with examples in Data Science

Predicted Values in Logistic Regression (Data Science – Exam Oriented Answer)

Logistic Regression is used when the dependent variable is **categorical** (usually binary: 0 or 1). Instead of predicting exact values like linear regression, it predicts **probabilities**.

1. How Predicted Values are Obtained

Step 1: Linear Combination (Logit)



First, input features are combined linearly:

z



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- z = logit (linear score)



1. How Predicted Values are Obtained

Step 1: Linear Combination (Logit)

First, input features are combined linearly:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

- z = logit (linear score)
- β_0 = intercept
- β_i = coefficients
- x_i = input variables

Step 2: Apply Logistic (Sigmoid) Function

$$p = \frac{1}{1+e^{-z}}$$



- Converts z into a probability (0 to 1)
- Converts the linear combination into a value between 0 and 1



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= input variables



Step 2: Apply Logistic (Sigmoid) Function

$$p = \frac{1}{1+e^{-z}}$$

- Converts z into a **probability** (0 to 1)
- Output p = predicted probability of class 1

Step 3: Convert Probability into Class Label

- If $p \geq 0.5 \rightarrow \text{Class} = 1$
- If $p < 0.5 \rightarrow \text{Class} = 0$

(Threshold can be changed based on problem)



2. Example in Data Science

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Suppose model equation is:



2. Example in Data Science

Example 1: Email Spam Detection

Suppose model equation is:

$$z = -2 + 0.8x$$

Where:

- x = number of suspicious words in email

Case: $x = 3$

$$z = -2 + 0.8(3) = 0.4$$

Apply sigmoid:

$$p = \frac{1}{1+e^{-0.4}} \approx 0.60$$

👉 **Prediction:**

- Probability = 0.60 ↓
- Since $0.60 > 0.5$, email = **Spam (1)**



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$$z = -2 + 0.8x$$

Where:

- x = number of suspicious words in email

Case: $x = 3$

$$z = -2 + 0.8(3) = 0.4$$

Apply sigmoid:

$$p = \frac{1}{1+e^{-0.4}} \approx 0.60$$

👉 **Prediction:**

- Probability = 0.60
- Since $0.60 > 0.5$, email = **Spam (1)**

Example 2: Student Pass/Fail Prediction

Model:

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$$z = -5 + 0.1 \times \text{marks}$$

Prediction: probability = 0.73

=



- student = **Pass** (1)

3. Key Points

- Logistic regression predicts **probabilities, not exact values**
- Uses **sigmoid function** to squash outputs between 0 and 1
- Final prediction depends on **decision threshold**
- Widely used in:
 - Spam detection
 - Disease prediction
 - Customer churn analysis

4. Conclusion

Predicted values in Logistic Regression are obtained by:



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2. Passing it through a **sigmoid function**

Interpretation of Logistic Regression Coefficients & Odds Ratio

◆ 1. Logistic Regression Model

Logistic regression predicts probability using the **logit function**:

$$\log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

- p = probability of event occurring
- LHS = **log-odds (logit)**
- β_i = coefficients

◆ 2. Interpretation of Coefficients (β)



 **Key Idea:**



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log-odds for a one-unit increase in the predictor variable.

• LHS = log-odds (logit)

=

= coefficients



◆ 2. Interpretation of Coefficients (β)

👉 Key Idea:

Each coefficient represents the **change in log-odds** for a **one-unit increase** in the predictor variable.

✓ Meaning:

- If $\beta_i > 0 \rightarrow$ probability **increases**
- If $\beta_i < 0 \rightarrow$ probability **decreases**
- If $\beta_i = 0 \rightarrow$ no effect

📌 Example:

Suppose:

$$\log \left(\frac{p}{1-p} \right) = -2 + \downarrow \times \text{StudyHours}$$

- Coefficient = 0.5



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◦ 1 extra hour of study $\rightarrow$ log-odds

increases by 0.5

predictor variable.

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- If $\beta_i < 0 \rightarrow$ probability **decreases**
- If $\beta_i = 0 \rightarrow$ no effect

Example:

Suppose:

$$\log \left(\frac{p}{1-p} \right) = -2 + 0.5 \times \text{StudyHours}$$

- Coefficient = **0.5**
- Interpretation:
 - 1 extra hour of study $\rightarrow$ **log-odds increases by 0.5**

But log-odds is not intuitive $\rightarrow$ we use **odds ratio**

◆ **3. Odds and Odds Ratio**



✓ **Odds:**



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• If $p = 0.8$ odds = 4 (4:1)

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◆ 3. Odds and Odds Ratio

✓ Odds:

$$\text{Odds} = \frac{p}{1-p}$$

- If $p = 0.8$, odds = 4 (4:1)

✓ Odds Ratio (OR):

$$\text{Odds Ratio} = e^{\beta_i}$$

- Converts coefficient into **multiplicative effect**

◆ 4. Interpretation of Odds Ratio



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Odds ratio tells how many times the odds
change for 1 unit increase in predictor



◆ 4. Interpretation of Odds Ratio

👉 Key Idea:

Odds ratio tells how many times the odds change for **1 unit increase in predictor**

✓ Cases:

Odds Ratio	Interpretation
OR > 1	Odds increase
OR < 1	Odds decrease
OR = 1	No effect

📌 Example:



If:

β



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$$OR = e^{0.5} \approx 1.65$$

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OR = 1 No effect

Example:

If:

$$\beta = 0.5$$

$$OR = e^{0.5} \approx 1.65$$

Interpretation:

- 1 unit increase in $x \rightarrow$ odds increase by **1.65 times (65% increase)**

Negative Coefficient Example:

If:



$$\beta = -0.7$$

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Interpretation:

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- 1 unit increase in $x \rightarrow$ odds increase by **1.65 times (65% increase)**

Negative Coefficient Example:

If:

$$\beta = -0.7$$

$$OR = e^{-0.7} \approx 0.50$$

Interpretation:

- Odds are **reduced by 50%**

◆ 5. Real-Life Example

Problem:



Predict whether a person will **buy a product**

(1/



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Model:



◆ 5. Real-Life Example

🎯 Problem:

Predict whether a person will **buy a product** (1/0)

Model:

$$\log \left(\frac{p}{1-p} \right) = -1 + 0.8 \times \text{Income}$$

✓ Interpretation:

- Coefficient = 0.8
- Odds Ratio:

$$e^{0.8} \approx 2.23$$

👉 Meaning:

- For every 1 unit increase in income:
 - Odds of buying  ease **2.23 times**



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◆ 6. Summary (Exam Points)



What is a Confusion Matrix?
Explain TP, FP, FN, TN with a
table in Data Science

Confusion Matrix (Data Science)

A **Confusion Matrix** is a performance evaluation tool used in **classification models**. It shows how well a model predicts actual classes by comparing **predicted values vs actual values**.

Confusion Matrix Table (Binary Classification)

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)



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Exp.

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Actual Negative	False Negative (FN)	True Negative (TN)

Explanation of Terms

1. True Positive (TP) ↓

- Model correctly predicts the positive



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- Example: Patient has disease and model predicts disease.

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- Example: Patient has disease and model predicts disease.

	Predicted Positive	Predicted Negative
Actual Positive (P)	True Positive (TP)	False Negative (FN)
Actual Negative (N)	False Positive (FP)	True Negative (TN)

Explanation of Terms

1. True Positive (TP)

- Model correctly predicts the positive class.
- Example: Patient has disease **and** model predicts disease.

2. True Negative (TN)

- Model correctly predicts the negative class.
- Example: Patient does not have disease **and** model predicts no disease.

3. False Positive (FP) (Type I Error)

- Model incorrectly predicts positive when it is actually negative.

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4. False Negative (FN) (Type II Error)

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- Model correctly predicts the negative class.
- Example: Patient does not have disease **and** model predicts no disease.

3. False Positive (FP) (Type I Error)

- Model incorrectly predicts positive when it is actually negative.
- Example: Patient does not have disease but model predicts disease.

4. False Negative (FN) (Type II Error)

- Model incorrectly predicts negative when it is actually positive.
- Example: Patient has disease but model predicts no disease.

Simple Example

Suppose I have a spam filter that is
is



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• TP → Spam email correctly detected

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Simple Example

Suppose a model predicts whether an email is spam:

- TP → Spam email correctly detected
- TN → Normal email correctly identified
- FP → Normal email wrongly marked as spam
- FN → Spam email missed (dangerous case)

Why Confusion Matrix is Important

- Helps calculate key metrics:
 - **Accuracy** = $(TP + TN) / \text{Total}$

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◦ F1-score

• TP → Spam email correctly detected

• TN → Normal email correctly identified

- FP → Normal email wrongly marked as spam
- FN → Spam email missed (dangerous case)

Why Confusion Matrix is Important

- Helps calculate key metrics:
 - **Accuracy** = $(TP + TN) / \text{Total}$
 - **Precision** = $TP / (TP + FP)$
 - **Recall** = $TP / (TP + FN)$
 - **F1-score**
- Gives deeper insight than accuracy alone, especially for **imbalanced datasets**

Conclusion (Exam Point)



A confusion matrix provides a structured way to evaluate a model's performance, helping to

derive important performance metrics and understand model behavior



Discuss F1-score, accuracy, and why accuracy can be misleading in Data science

1. Accuracy

Definition:

Accuracy measures the proportion of correctly classified instances out of total instances.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

Where:

- **TP (True Positive):** Correctly predicted positive cases
- **TN (True Negative):** Correctly predicted negative cases
- **FP (False Positive):** Incorrectly predicted positive
- **FN (False Negative):** Incorrectly predicted negative



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Example.

If out of 100 samples, 90 are correctly

• TP (True Positive): Correctly predicted

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- **FP (False Positive):** Incorrectly predicted positive
- **FN (False Negative):** Incorrectly predicted negative

Example:

If out of 100 samples, 90 are correctly predicted →

Accuracy = $90/100 = 90\%$

2. F1-Score

Definition:

F1-score is the **harmonic mean** of Precision and Recall. It balances both.

- **Precision** = $TP / (TP + FP)$ → How many predicted positives are correct
- **Recall** = $TP / (TP + FN)$ → How many actual positives are captured



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Key Point:



2. F1-Score

Definition:

F1-score is the **harmonic mean** of Precision and Recall. It balances both.

- **Precision** = $TP / (TP + FP)$ → How many predicted positives are correct
- **Recall** = $TP / (TP + FN)$ → How many actual positives are captured

$$F1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Key Point:

- F1-score is high only when **both Precision and Recall are high**
- Useful when **class distribution is imbalanced**

Example:

Precision = 0.8, Recall = 0.6

F1-score =

$$2 \times (0.8 \times 0.6) / (0.8 + 0.6) = 0.685$$



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3. Why Accuracy Can Be Misleading

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and Recall are high

- Useful when **class distribution is imbalanced**

Example:

Precision = 0.8, Recall = 0.6

F1-score =

$$2 \times (0.8 \times 0.6) / (0.8 + 0.6) = 0.685$$

3. Why Accuracy Can Be Misleading

Accuracy works well only when **classes are balanced**. It fails in **imbalanced datasets**.

● Problem Scenario (Imbalanced Data)

Suppose:

- 1000 total samples
- 950 = Negative class
- 50 = Positive class



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- TP = 0, TN = 950, FP = 0, FN = 50

=



3. Why Accuracy Can Be Misleading

Accuracy works well only when **classes are balanced**. It fails in **imbalanced datasets**.

● Problem Scenario (Imbalanced Data)

Suppose:

- 1000 total samples
- 950 = Negative class
- 50 = Positive class

A model predicts **all as Negative**:

- TP = 0, TN = 950, FP = 0, FN = 50

Accuracy:

$$= \frac{950}{1000} = 95\%$$

👉 Looks excellent, but:



- Model **fails completely** to detect positive cases



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Accuracy:

=

$$= \frac{950}{1000} = 95\%$$



👉 Looks excellent, but:

- Model **fails completely** to detect positive cases
- Recall = 0 → Very poor model

4. Why F1-Score is Better in Such Cases

- F1 considers both **False Positives** and **False Negatives**
- Penalizes models that ignore minority class
- Gives a **more realistic performance measure**

👉 In above example:

- Precision = 0
- Recall = 0
- F1-score = 0 → Correctly shows bad



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4. Why F1-Score is Better in Such Cases

- F1 considers both **False Positives** and **False Negatives**
- Penalizes models that ignore minority class
- Gives a **more realistic performance measure**

👉 In above example:

- Precision = 0
- Recall = 0
- F1-score = 0 → Correctly shows **bad model**

5. Summary (Exam-Oriented)



Metric	Formula	Use
Precision	$TP / (TP + FP)$	When you want to know how many of the predicted positive cases are actually positive



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Explain ROC Curve. How is it used to evaluate classification models in Data Science

ROC Curve (Receiver Operating Characteristic Curve)

The **ROC Curve** is a graphical tool used to evaluate the performance of a **binary classification model** at different classification thresholds.

It shows the trade-off between:

- **True Positive Rate (TPR)** → Sensitivity / Recall
- **False Positive Rate (FPR)** → Fall-out

Key Metrics Used in ROC



1. True Positive Rate (TPR) / Sensitivity

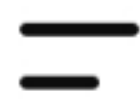
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2. False Positive Rate (FPR)



Key Metrics Used in ROC

1. True Positive Rate (TPR) / Sensitivity

$$TPR = \frac{TP}{TP+FN}$$

2. False Positive Rate (FPR)

$$FPR = \frac{FP}{FP+TN}$$

ROC Curve Representation

$$y = TPR \quad \text{vs} \quad x = FPR$$

- X-axis → **False Positive Rate (FPR)**
- Y-axis → **True Positive Rate (TPR)**
- Each point represents a different **threshold value**



How ROC Curve Works



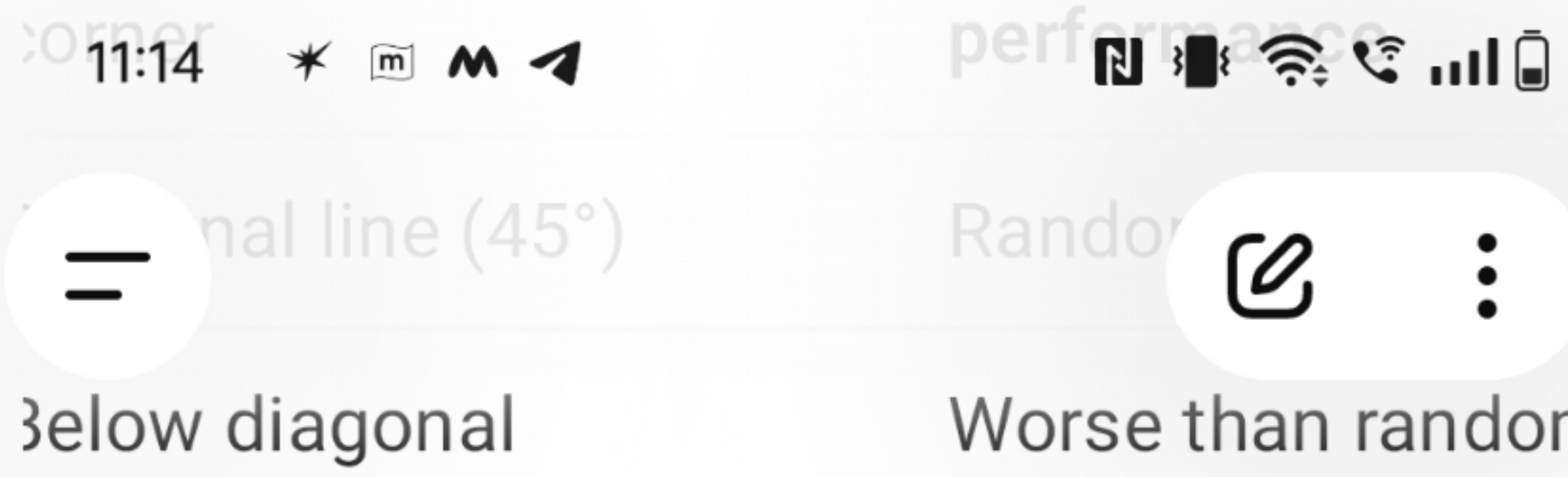
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probabilities



- **AUC** measures the entire 2D area under the ROC curve



AUC (Area Under Curve)

- **AUC** measures the entire 2D area under the ROC curve
- Range: **0 to 1**

AUC Value	Interpretation
1.0	Perfect model
0.9 – 0.99	Excellent
0.8 – 0.9	Good
0.7 – 0.8	Fair
0.5	Random

How ROC is Used to Evaluate Models

1. **+** Reply to ChatGPT  
- Compare multiple models using AUC
- Higher AUC $\rightarrow$ Better classifier



How ROC is Used to Evaluate Models

1. Model Comparison

- Compare multiple models using AUC
- Higher AUC → Better classifier

2. Threshold Selection

- Helps choose optimal threshold based on:
 - ▶ High recall (TPR)
 - ▶ Low false positives (FPR)

3. Class Imbalance Handling

- Works better than accuracy when dataset is imbalanced

4. Performance Visualization

- Shows trade-off between sensitivity and specificity clearly



Example



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Suppose a disease detection model.

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- Shows trade-off between sensitivity and specificity clearly

Example

Suppose a disease detection model:

- Threshold = 0.5 → moderate TPR & FPR
- Threshold = 0.2 → high TPR but also high FPR
- Threshold = 0.8 → low FPR but may miss positives

ROC helps select the **best balance**.

Advantages



- Threshold-independent evaluation



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Advantages

- Threshold-independent evaluation
- Useful for comparing classifiers
- Works well for imbalanced datasets

Limitations

- Less informative when classes are highly skewed (Precision-Recall curve may be better)
- Does not directly reflect real-world cost of errors



Conclusion (Exam Ready)



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The ROC curve is a plot that plots TPR vs FPR across thresholds. It helps in model comparison, threshold tuning,



Compare Logistic Regression with Linear Regression

Fast answer >

Logistic regression and linear regression are both supervised learning algorithms, but they are used for different types of problems and behave quite differently.

1. Type of problem

- **Linear regression** → used for **regression problems** (predicting continuous values like price, temperature).
- **Logistic regression** → used for **classification problems** (predicting categories like yes/no, spam/not spam).

2. Output

- **Linear regression** outputs any real value ($-\infty$ to $+\infty$).



- **Logistic regression** outputs a probability (0 to 1), which is then typically converted into a class label.



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like price, temperature).

- **Logistic regression** → used for **classification problems** (predicting categories like yes/no, spam/not spam).

2. Output

- **Linear regression** outputs **any real value** ($-\infty$ to $+\infty$).
- **Logistic regression** outputs a **probability** (**0 to 1**) using a sigmoid function, which is then typically converted into a class label.

3. Model equation

- Linear regression:

$$y = \beta_0 + \beta_1 x + \dots + \beta_n x_n$$

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x + \dots + \beta_n x_n)}}$$



4. Function used



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- **Logistic regression** → **sigmoid** (S-shaped

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- **Logistic regression** outputs a **probability** (0 to 1) using a sigmoid function, which is then typically converted into a class label.

3. Model equation

- Linear regression:

$$y = \beta_0 + \beta_1 x + \dots + \beta_n x_n$$

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x + \dots + \beta_n x_n)}}$$

4. Function used

- **Linear regression** → identity function (straight line).
- **Logistic regression** → **sigmoid (S-shaped curve)** to squash outputs into probabilities.

5. Loss function



- **Linear regression** → Mean Squared Error



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- **Logistic regression** → **Log Loss** (Cross-Entropy).

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$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x + \dots + \beta_n x_n)}}$$

4. Function used

- **Linear regression** → identity function (straight line).
- **Logistic regression** → **sigmoid (S-shaped curve)** to squash outputs into probabilities.

5. Loss function

- **Linear regression** → Mean Squared Error (MSE).
- **Logistic regression** → Log Loss (Cross-Entropy).

6. Interpretation

- **Linear regression** predicts exact values.
- **Logistic regression** ↓ predicts the probability of belonging to a class.

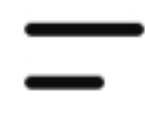
7.



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- **Linear regression** has no decision



Loss function



- **Linear regression** → Mean Squared Error (MSE).
- **Logistic regression** → Log Loss (Cross-Entropy).

6. Interpretation

- **Linear regression** predicts exact values.
- **Logistic regression** predicts the probability of belonging to a class.

7. Decision boundary

- **Linear regression** has no decision boundary.
- **Logistic regression** uses a **threshold** (e.g., 0.5) to classify outputs into categories.

In short:

Linear regression predicts *how much*, while logistic regression predicts *which class*.



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